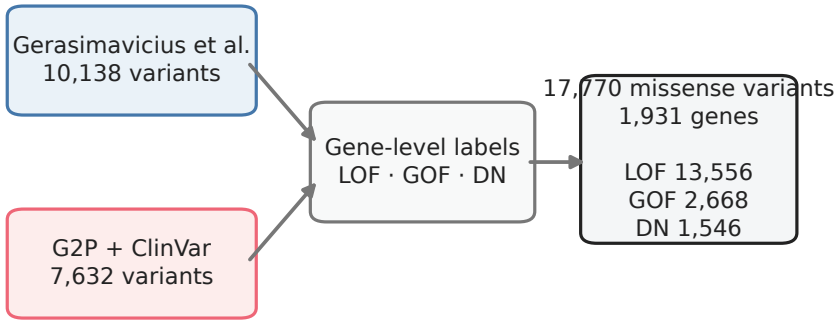
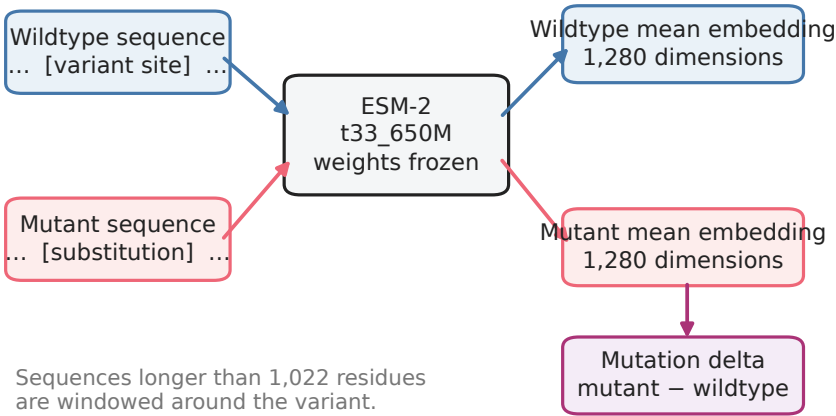


A Mechanism cohort



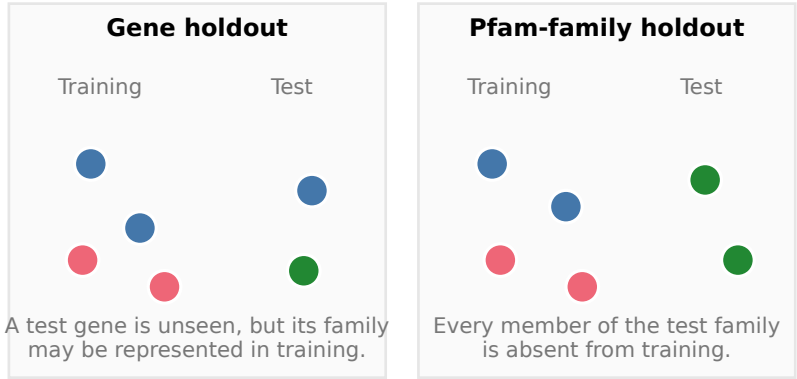
Each missense variant inherits its gene's curated mechanism label.

B Frozen ESM-2 representations



Sequences longer than 1,022 residues are windowed around the variant.

C Held-out unit changes the transfer question



D Shared evaluation framework

Biological target	Representation	Held-out unit
Disease mechanism	WT · mutant · delta	Gene · Pfam family
Pathogenicity	Delta	Gene · Pfam family
Enzyme type	Wildtype	Gene · Pfam family
Folding stability	Delta	Random · domain · Pfam family

Frozen embeddings, task-specific probes, and uncertainty estimates matched to the held-out unit.